

Saving the Rainbow of Life

II PUC · Ch 13 · Biodiversity and Conservation

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



The Rainbow of Life

"**Biodiversity** is the variety of life," said Prof. BAIO, quoting Edward Wilson. "It has three levels: **genetic diversity** — like the many reserpine-making forms of Rauwolfia; **species diversity** — the Western Ghats have more amphibians than the Eastern Ghats; and **ecological diversity** — deserts, rain forests, mangroves, coral reefs."

💡 **REMEMBER**

India has more than 50,000 varieties of rice and about 1,000 of mango.



How Many Species?

About 1.5 million species have been described, but Robert May estimated there may be about 7 million. More than 70% of all animals are insects! India has only 2.4% of the world's land but 8.1% of its species — one of the 12 **mega-diversity** countries.

💡 **REMEMBER**

Plants (including algae and fungi) make up no less than 22% of all recorded species.



Patterns on the Map

Diversity falls as you go from the equator to the poles — the **latitudinal gradient**. Colombia has about 1,400 bird species; Greenland only 56. The **Amazon rain forest** is the richest of all, because the tropics are old, stable and get lots of sunlight.

💡 **REMEMBER**

Tropical environments are less seasonal and more predictable.



More Area, More Species

Explorer **Alexander von Humboldt** found that species richness rises with area — the **species–area relationship**: log S = log C + Z log A. For most areas Z is 0.1 to 0.2; for whole continents it is 0.6 to 1.2.

💡 **REMEMBER**

On a log–log graph the relationship is a straight line.



Why It Matters

Paul Ehrlich compared species to the rivets of an aeroplane — the **rivet popper hypothesis**. Lose a few and it still flies; lose key ones and it crashes. David Tilman found that plots with more species gave more stable yields year after year.

💡 **REMEMBER**

Reasons to conserve: narrowly utilitarian, broadly utilitarian and ethical.



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The Evil Quartet

Four villains cause species loss — the **evil quartet**: **habitat loss** and fragmentation; **over-exploitation** (Steller's sea cow, passenger pigeon); **alien species** like the Nile perch in Lake Victoria, Parthenium and Lantana; and **co-extinctions**, when a species dies with its host.

💡 REMEMBER

The Nile perch wiped out more than 200 species of cichlid fish.



Saving at Home

In situ conservation protects species in their natural homes. There are 34 **biodiversity hotspots**, three in India: Western Ghats–Sri Lanka, Indo-Burma and the Himalaya. India also has biosphere reserves, national parks, wildlife sanctuaries and **sacred groves** protected by local communities.

💡 **REMEMBER**

Hotspots cover less than 2% of Earth's land area.



Saving Away from Home

Ex situ conservation protects species outside their homes — in zoos, botanical gardens and wildlife safari parks. Gametes can be stored for long periods by **cryopreservation**, and seeds in **seed banks**. Countries promised to work together at the Earth Summit in Rio (1992) and the World Summit in Johannesburg (2002).

💡 **REMEMBER**

Bindu's promise: "I will protect the rainbow of life."



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

Biodiversity p.1 · genetic diversity p.1 · species diversity p.1 · ecological diversity p.1 · mega-diversity p.2 · latitudinal gradient p.3 · Amazon rain forest p.3 · Alexander von Humboldt p.4 · species–area relationship p.4 · rivet popper hypothesis p.5 · evil quartet p.6 · habitat loss p.6 · over-exploitation p.6 · alien species p.6 · co-extinctions p.6 · In situ conservation p.7 · biodiversity hotspots p.7 · sacred groves p.7 · Ex situ conservation p.8 · cryopreservation p.8 · seed banks p.8



QUICK QUIZ

Do you remember?

1. The term biodiversity was popularised by:

- Paul Ehrlich
- Edward Wilson
- Humboldt
- Tilman

2. The rivet popper hypothesis was given by:

- Paul Ehrlich
- Robert May
- David Tilman
- Gause

3. For most regions, the value of Z is:

- 0.1–0.2
- 0.6–1.2
- 1–2
- 5–10

4. The Nile perch is an example of:

- Over-exploitation
- Alien species invasion
- Co-extinction
- Habitat loss

5. Number of biodiversity hotspots in India:

- 1
- 2
- 3
- 34

Answers: 1-Edward Wilson · 2-Paul Ehrlich · 3-0.1–0.2 · 4-Alien species invasion · 5-3